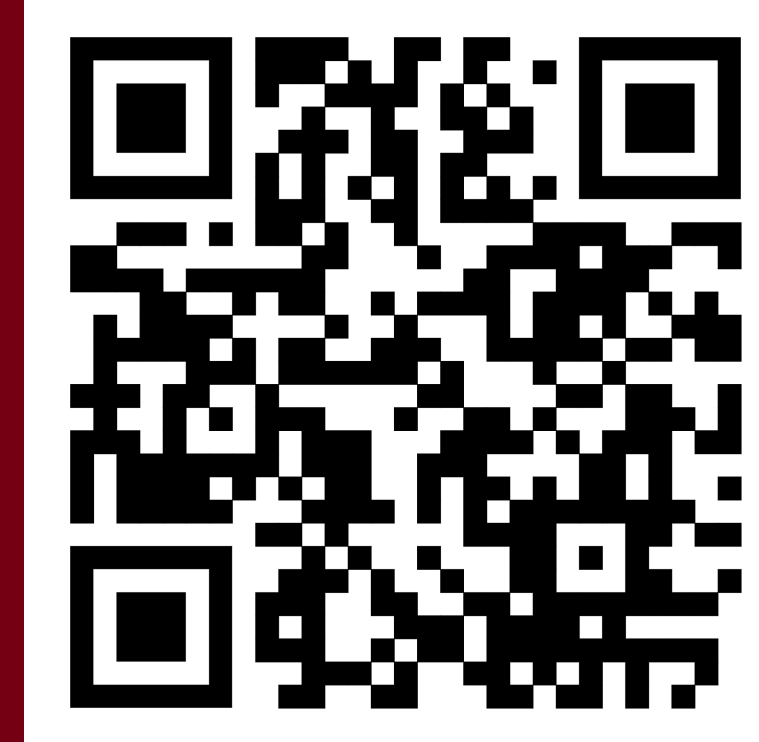




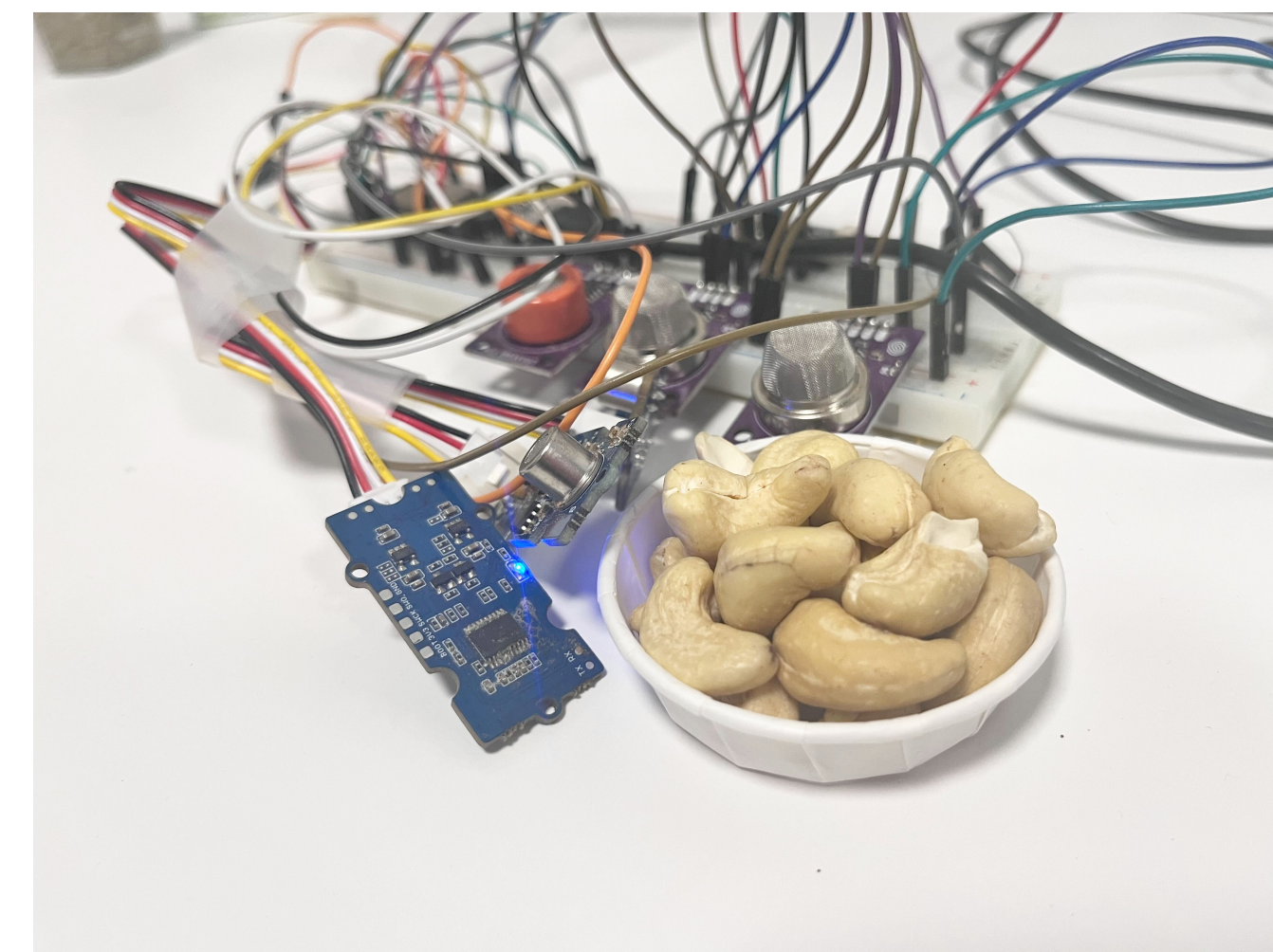
Demo



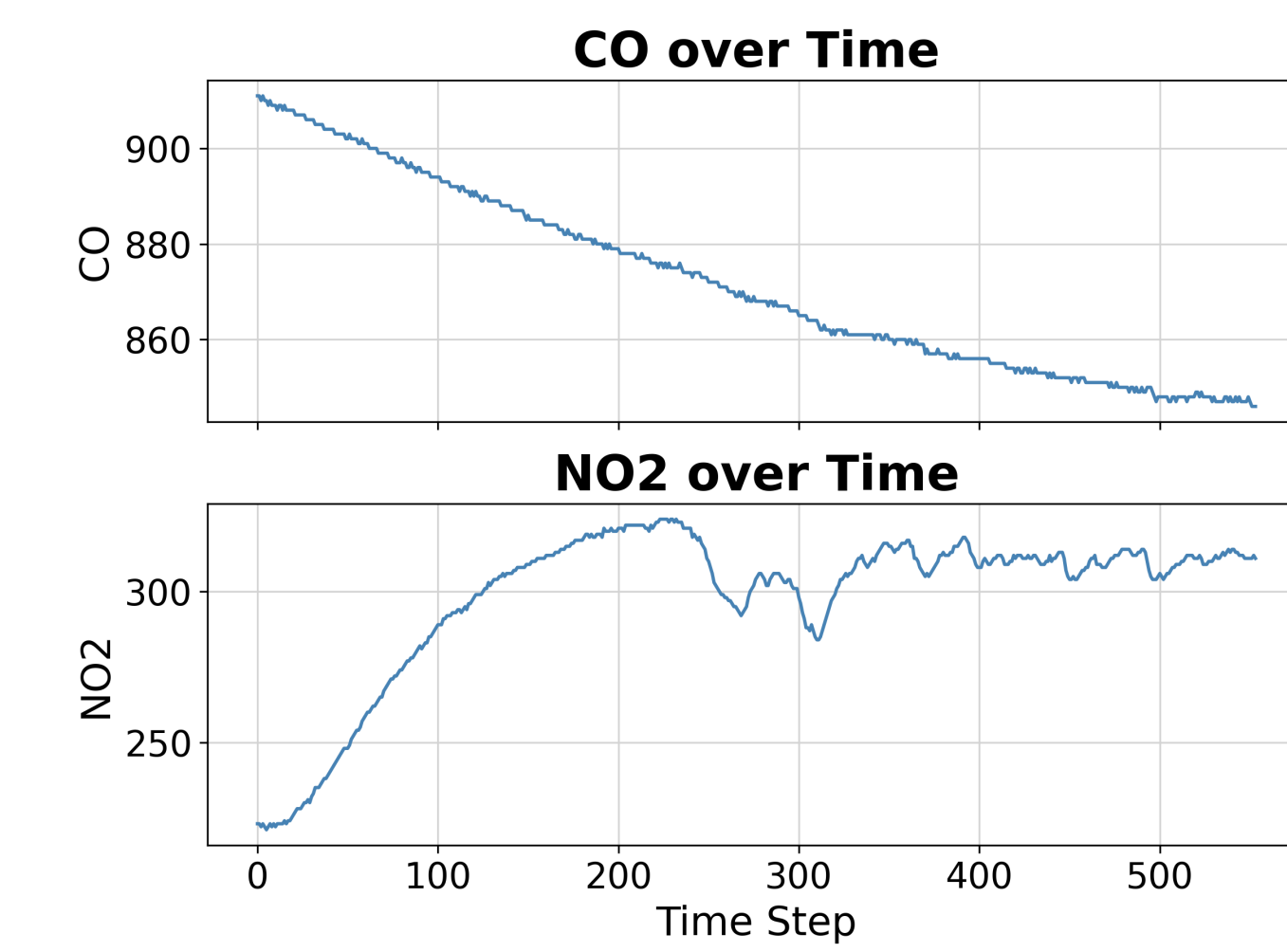
Paper / Code

Can AI recognize smell from low-cost portable sensors?

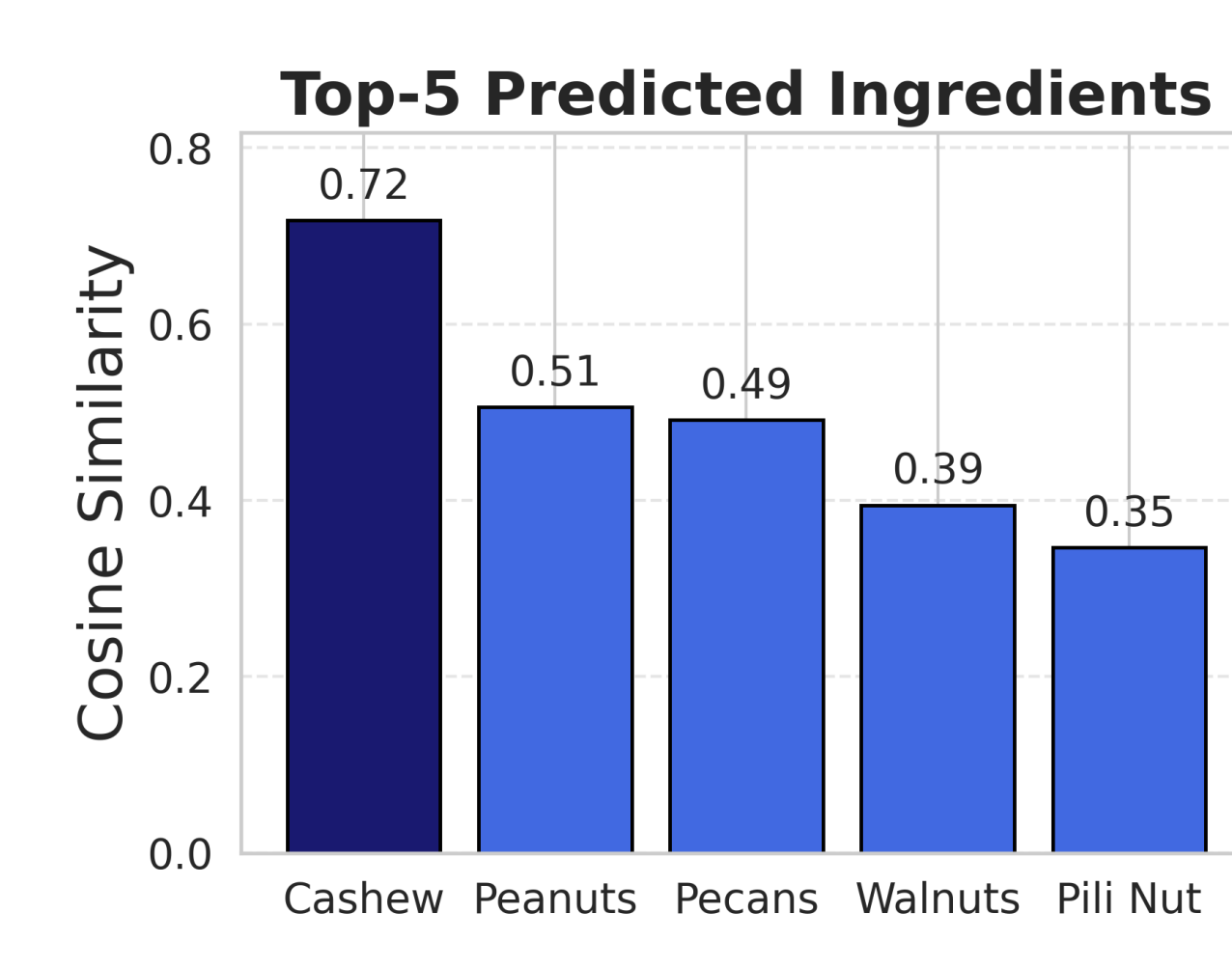
We introduce **SmellNet**, a large benchmark of multichannel MOX (metal-oxide) sensor time series for both **single-substance recognition** and **mixture prediction**, and **ScentFormer**, a temporal model with optional training-time GC-MS supervision.



(a) Portable sensor setup detecting cashew.

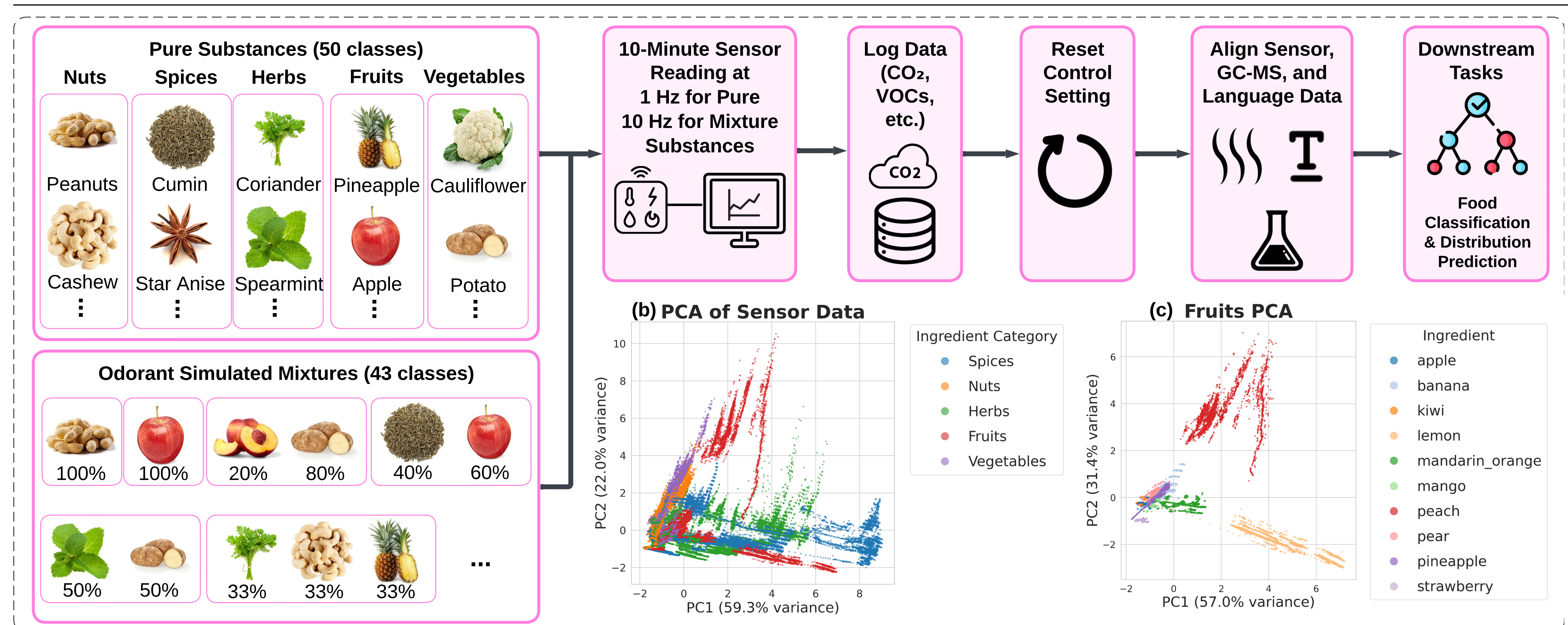


(b) Example multichannel sensor responses over time.



(c) Example top-5 predictions from the model.

Data Collection Pipeline



Controlled sensing of natural foods and mixtures. Repeated sensor recordings are aligned with metadata and optional ingredient-level GC-MS priors. PCA shows broad category structure, but substantial overlap remains, making ingredient recognition and unseen-mixture generalization challenging.

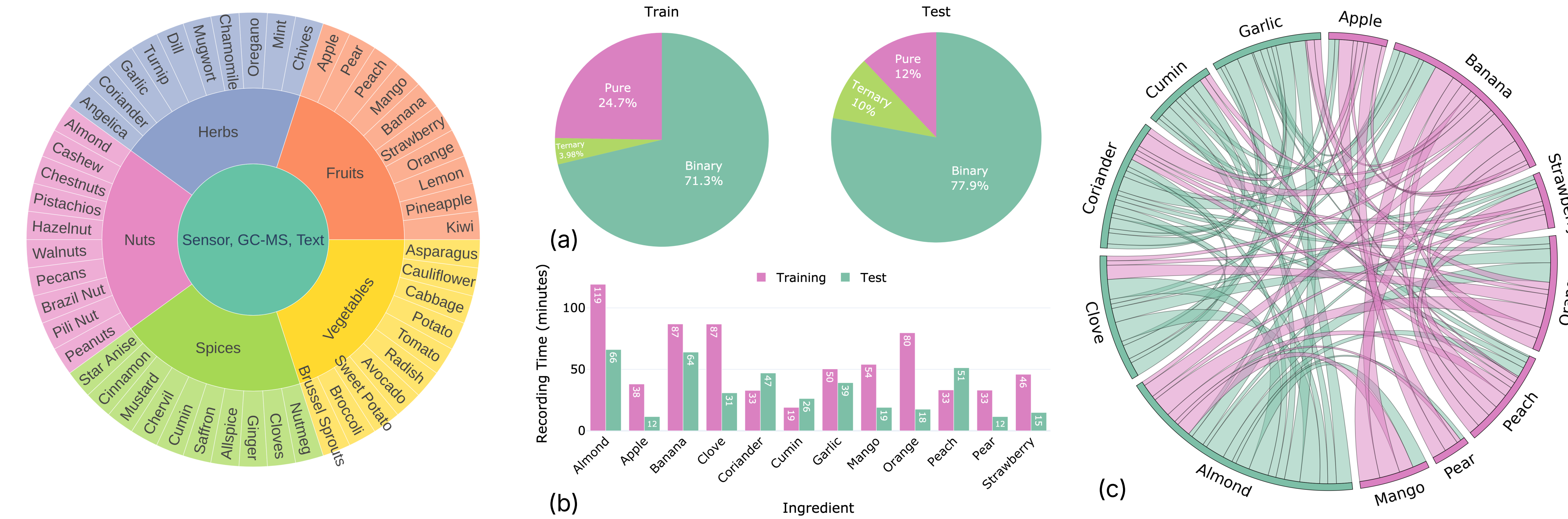
SMELLNET at a Glance

828K timesteps **50** substances **43** mixtures **68** hours

A large-scale portable MOX benchmark for both ingredient classification and mixture prediction.

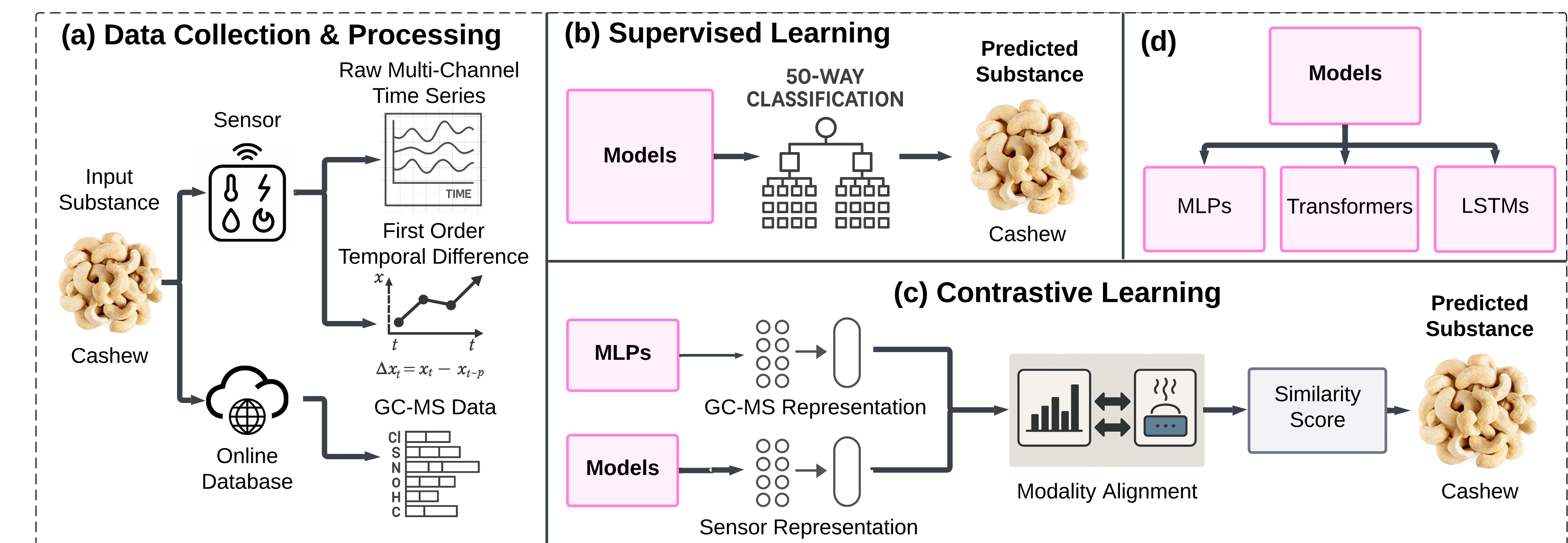
Why this benchmark matters. Prior smell datasets are often small, non-portable, or not designed for real-time sensor learning; **SmellNet** provides a larger benchmark for practical machine olfaction.

Dataset Structure



SmellNet-Base is a 50-way classification task, and **SmellNet-Mixture** is for predicting mixture ratio. The mixture benchmark includes **seen** and **unseen** splits for session robustness and compositional generalization.

From Sensor Signals to Predictions



Method overview. Raw multichannel sensor streams are optionally differenced, segmented into sliding windows, and encoded for classification or mixture prediction. During training, sensor embeddings can be aligned with ingredient-level GC-MS embeddings. **No new GC-MS is required at inference.**

Key Design Choices

Preprocessing

Temporal differencing reduces drift and highlights response dynamics:

$$\Delta x_t = x_t - x_{t-p}$$

Sliding windows improve data efficiency.

ScentFormer

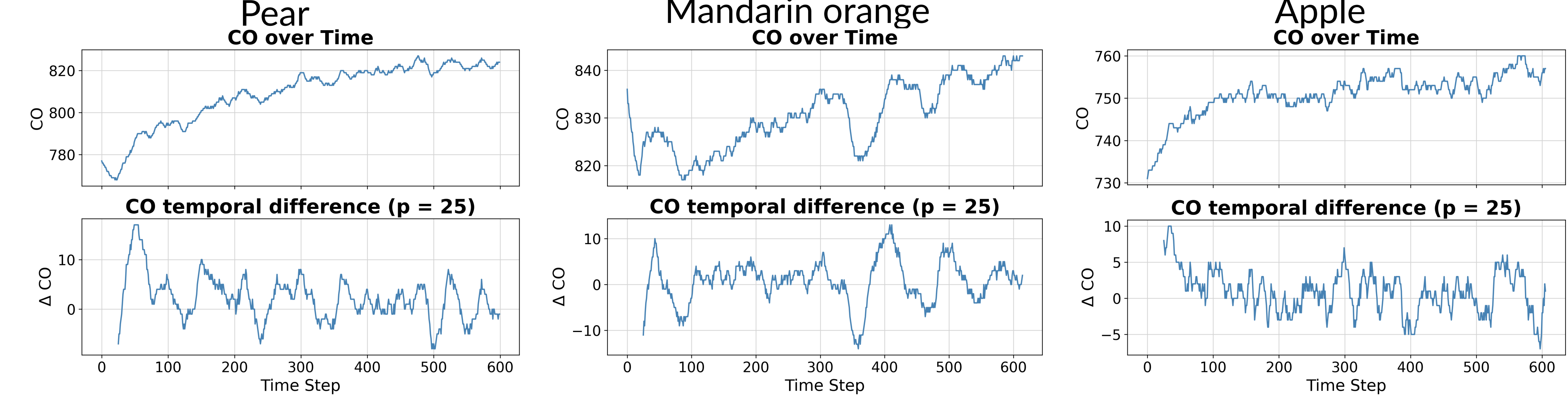
A pre-norm Transformer encoder processes windowed multichannel sensor sequences and pools them into a fixed representation for downstream prediction.

Outputs

Base: 50-way classification
Mixture: 12-D normalized ratio prediction
GC-MS: ingredient-level aligned label space

Why it works: Temporal dynamics reduce drift effects. GC-MS provides chemistry-aware training supervision.

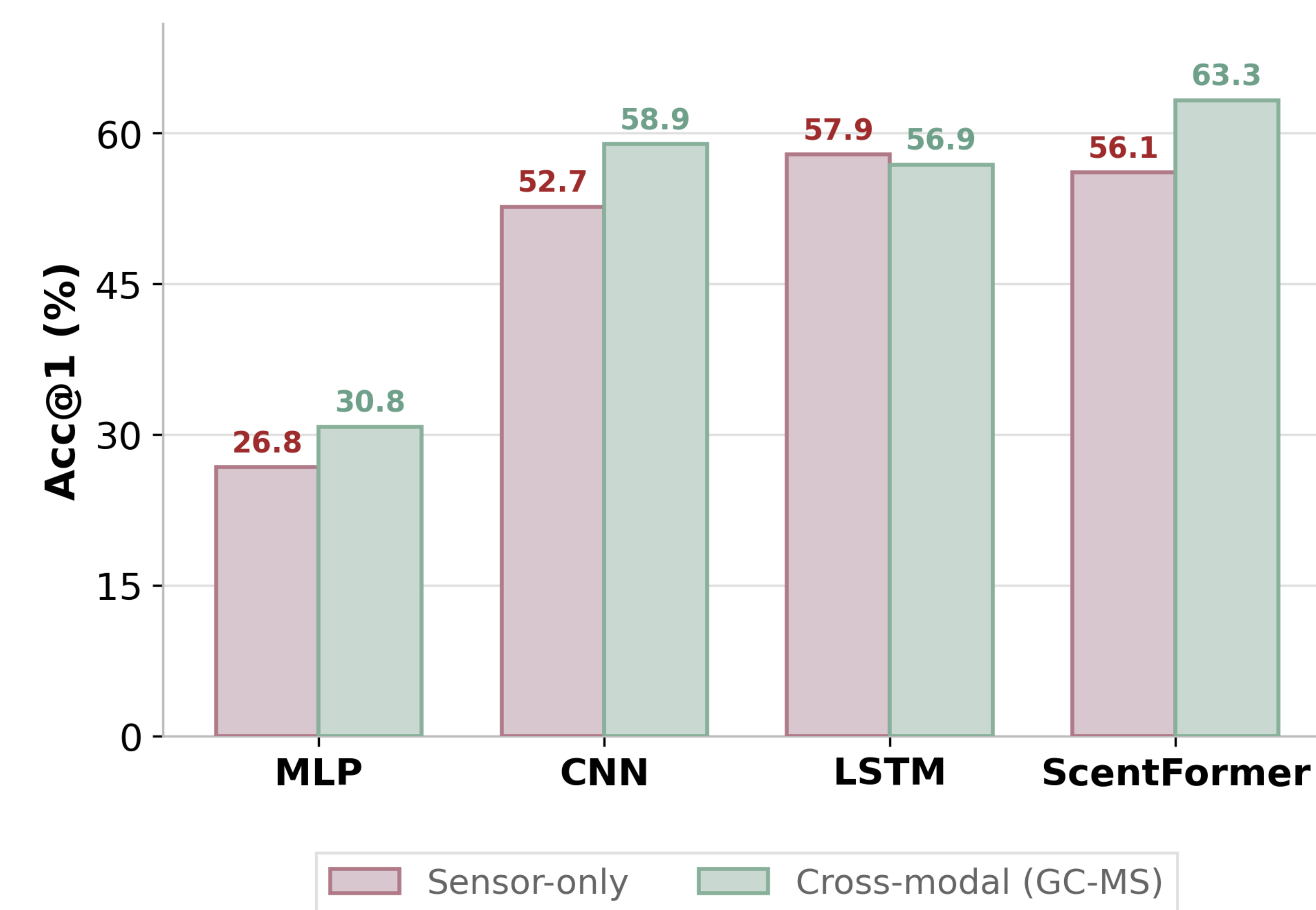
Why Temporal Differencing Helps



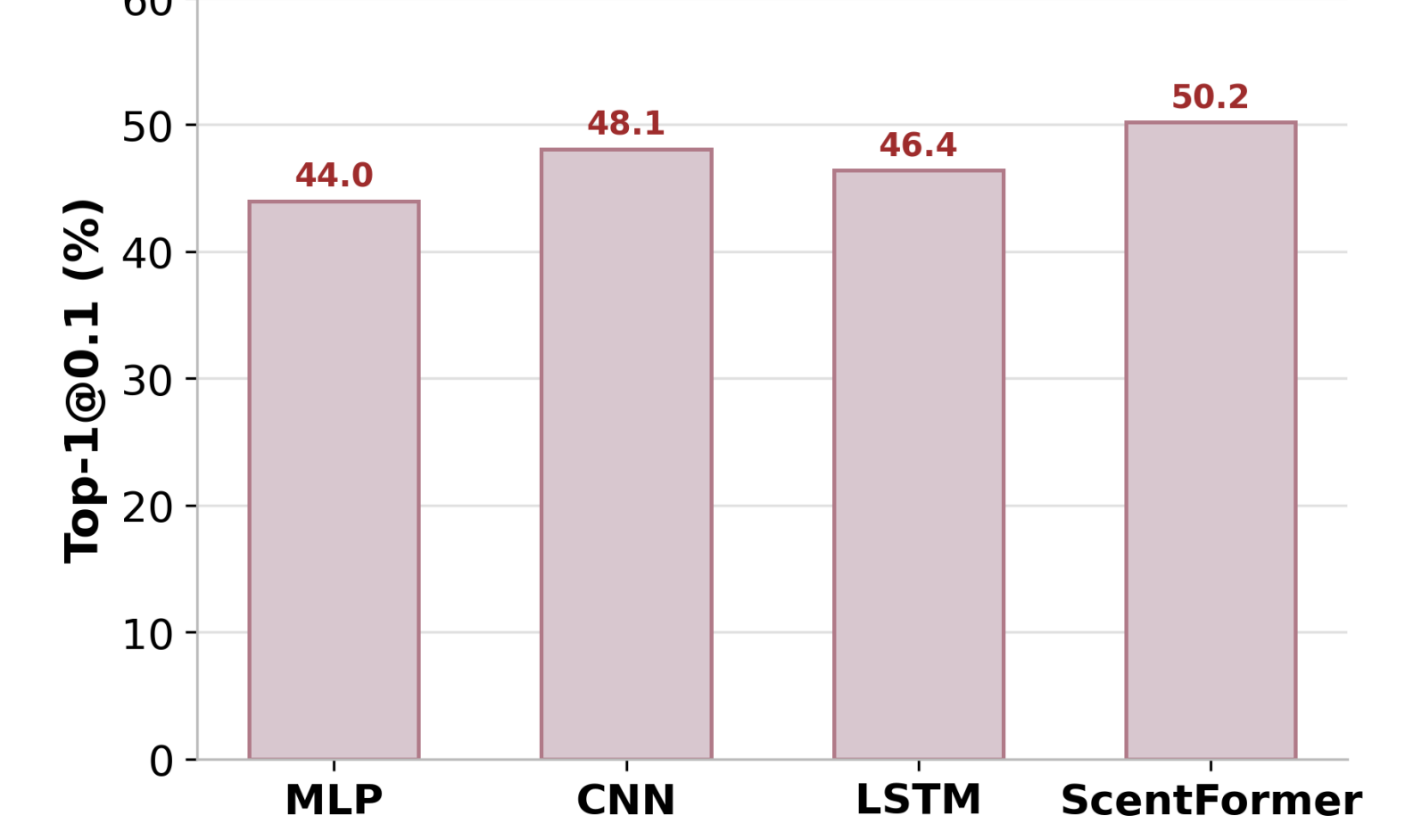
Temporal differencing suppresses drift and baseline shifts, exposing transient odor-response patterns that are easier to distinguish.

Results

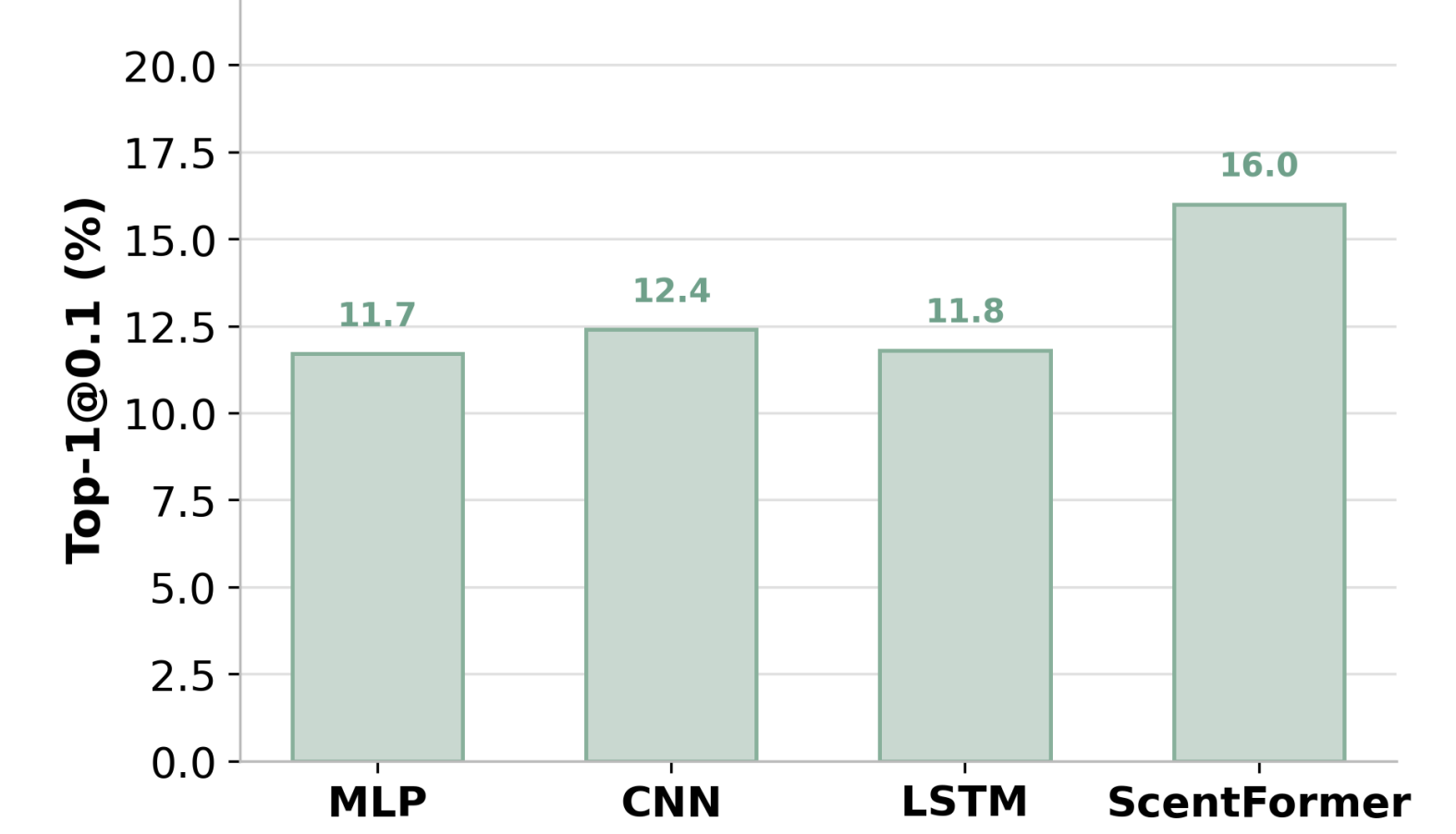
Main Results on SmellNet-Base



Seen Mixtures



Unseen Mixtures



Temporal modeling improves smell recognition, and **ScentFormer** performs best overall. On **SmellNet-Base**, ScentFormer reaches **63.3% Top-1** with GC-MS supervision. On **SmellNet-Mixture**, it reaches **50.2% Top-1@0.1** on **seen** mixtures but only **16.0%** on **unseen mixtures**, showing that compositional generalization remains the main bottleneck.

Takeaway

Portable MOX sensors enable meaningful smell recognition in controlled settings, but unseen-mixture generalization remains the key challenge.

Limitations & Next Steps

Performance may drop in new environments because data were collected in controlled settings, and low-cost MOX signals are noisy and low-resolution. Future work includes broader data collection, stronger compositional training, and improved sensors or multimodal supervision.